Examples: e^{2x}, e^{x^2+1} and 2^x are exponential functions.

5.1.20 Logarithmic function

For $x > 0$, $a > 0$ and $a \neq 1$, a function $f(x)$ defined by $f(x) = \log_a x$ is called the logarithmic function.

Remark Domain is $(0, \infty)$, range is R and $(1, 0)$ is a point on the graph

Example: $f(x) = \log_e(x+2)$ and $f(x) = \log_e(\sin x)$ are logarithmic functions.

5.1.21 Sum, difference, product and quotient of two functions

If $f(x)$ and $g(x)$ are two functions having same domain and co-domain, then

(i) $(f \pm g)(x) = f(x) \pm g(x)$

(ii) $(fg)(x) = f(x)\, g(x)$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$

(iv) $(cf)(x) = cf(x)$, c is a constant.

5.1.22 Graph of a function

The graph of a function is the set of points $(x, f(x))$ where x belongs to the domain of the function and $f(x)$ is the value of the function at x.

To draw the graph of a function, we find a sufficient number of ordered pairs $(x, f(x))$ belonging to the function and join them by a smooth curve.

NOTE

It is enough to draw a diagram for graph of a function in white sheet itself.

Example 5.1

Find the domain for which the functions $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal.

Solution

Given that $f(x) = g(x)$

$\Rightarrow \quad 2x^2 - 1 = 1 - 3x$

$2x^2 + 3x - 2 = 0$

$(x+2)(2x-1) = 0$

$\therefore \; x = -2, \; x = \frac{1}{2}$

Domain is $\left\{-2, \frac{1}{2}\right\}$

Example 5.2

$f = \{(1, 1), (2, 3)\}$ be a function described by the formula $f(x) = ax+b$. Determine a and b ?

Solution

Given that $f(x) = ax + b$, $f(1) = 1$ and $f(2) = 3$

$\therefore \quad a + b = 1$ and $2a + b = 3$

$\Rightarrow \quad a = 2$ and $b = -1$

Example 5.3

If $f(x) = x + \frac{1}{x}$, $x > 0$, then show that $[f(x)]^3 = f(x^3) + 3f\left(\frac{1}{x}\right)$

Solution

$f(x) = x + \frac{1}{x}$, $f(x^3) = x^3 + \frac{1}{x^3}$ and $f\left(\frac{1}{x}\right) = f(x)$

Now, $LHS = [f(x)]^3$

$= \left[x + \frac{1}{x}\right]^3$

$= x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$

$= f(x^3) + 3f(x)$

$= f(x^3) + 3f\left(\frac{1}{x}\right)$

$= RHS$

Example 5.4

Let f be defined by $f(x) = x - 5$ and g be defined by

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